

7-5

正定

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半正定

An $n \times n$ matrix C is said to be positive definite if C is symmetric and $v^T C v > 0$ for every nonzero vector v in R^n . We say C is positive semidefinite if C is symmetric and $v^T C v \geq 0$ for every nonzero vector v in R^n .



$v \neq 0$
 $\boxed{\text{result}} > 0$
 $\boxed{x}$

$v^T C v > 0, v \neq 0$

Consider $C = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$

$\begin{bmatrix} a & b \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} \quad a \neq 0, b \neq 0$

$= \begin{bmatrix} a-b & a+b \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = a^2 + b^2 > 0$

Application: <https://youtu.be/QW6uINn7uGk>

An $n \times n$ matrix C is said to be positive definite if C is symmetric and $\mathbf{v}^T C \mathbf{v} > 0$ for every nonzero vector $\mathbf{v}$ in R^n . We say C is positive semidefinite if C is symmetric and $\mathbf{v}^T C \mathbf{v} \geq 0$ for every nonzero vector $\mathbf{v}$ in R^n .

$\boxed{A^T = A}$ $A^T = (Q B Q^T)^T = Q B^T Q^T = Q B Q^T = A$

66. Suppose that $\underline{A} = Q B Q^T$, where Q is an orthogonal matrix and B is positive definite. Prove that A is positive definite.

$\boxed{v \neq 0}$ $v^T B v > 0, v \neq 0$ $(Q^T v)^T = v^T Q$ $w \neq 0$ $w \neq 0$

$v^T A v = \underline{v^T Q B Q^T v}$ $v \neq 0 \Rightarrow Q^T v \neq 0$ $w \neq 0$

$= \underbrace{(Q^T v)^T}_w B \underbrace{(Q^T v)}_w = w^T B w > 0$

59. Suppose that A is a symmetric matrix. Prove that A is positive definite if and only if all of its eigenvalues are positive.

$$\Rightarrow: \quad v \neq 0, \quad v^T A v > 0$$

$$A v = \lambda v$$

$$\begin{aligned} v^T A v \\ = v \cdot (A v) \end{aligned}$$

$$v \cdot (A v) > 0$$

$$\Rightarrow v \cdot (\lambda v) > 0 \Rightarrow \lambda \|v\|^2 > 0 \Rightarrow \lambda > 0$$

How about non-symmetric?

59. ~~Suppose that A is a symmetric matrix.~~ Prove that A is positive definite if and only if all of its eigenvalues are positive.

$\Leftarrow : A = P D P^T$

$\downarrow$

$\begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}$

$\downarrow$

positive.

D positive definite.

$$\begin{bmatrix} v_1 & \dots & v_n \end{bmatrix} \begin{bmatrix} d_1 & & \\ & \ddots & \\ & & d_n \end{bmatrix} \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}$$
$$= d_1 v_1^2 + d_2 v_2^2 + \dots + d_n v_n^2 > 0$$

$\Rightarrow A$ is positive definite.

$$v^T A v > 0$$

$v \neq 0$

61. Suppose that A is invertible and positive definite. Prove that A^{-1} is positive definite.

$$v^T A^{-1} v > 0 \quad ?$$

$$\lambda_1, \lambda_2, \dots, \lambda_n > 0$$

A eigenvalue

$$A^{-1} \text{ eigenvalue}$$

$$1/\lambda_1, 1/\lambda_2, \dots, 1/\lambda_n > 0$$

positive definite.

72. Prove that, for any matrix A , the matrices $A^T A$ and AA^T are positive semidefinite.

$$v \neq 0, \quad v^T A^T A v \geq 0 \quad \text{goal.}$$

73. Prove that, for any invertible matrix A , the matrices $A^T A$ and AA^T are positive definite.

$$v \neq 0, \quad v^T A^T A v > 0 \quad \text{goal.}$$

$$v \neq 0$$

$$v^T A^T A v = (Av)^T Av = Av \cdot Av$$

$$= \|Av\|^2 \geq 0$$

$$\text{If } A \text{ is inv.} \quad Av \neq 0 \quad \|Av\|^2 > 0$$

Spectral Decomposition

Orthonormal basis

$$A = PDP^T \quad \text{Let } P = [\mathbf{u}_1 \ \mathbf{u}_2 \ \cdots \ \mathbf{u}_n] \text{ and } D = \text{diag}[\lambda_1 \ \lambda_2 \ \cdots \ \lambda_n].$$

$$= P[\lambda_1 \mathbf{e}_1 \ \lambda_2 \mathbf{e}_2 \ \cdots \ \lambda_n \mathbf{e}_n]P^T$$

$$= [\lambda_1 P\mathbf{e}_1 \ \lambda_2 P\mathbf{e}_2 \ \cdots \ \lambda_n P\mathbf{e}_n]P^T$$

$$= [\lambda_1 \mathbf{u}_1 \ \lambda_2 \mathbf{u}_2 \ \cdots \ \lambda_n \mathbf{u}_n] \begin{bmatrix} \mathbf{u}_1^T \\ \mathbf{u}_2^T \\ \vdots \\ \mathbf{u}_n^T \end{bmatrix}$$

$n \times 1$
 $1 \times n$




$$= \lambda_1 P_1 + \lambda_2 P_2 + \cdots + \lambda_n P_n$$

P_i are symmetric

Let A be an $n \times n$ symmetric matrix.

53. Use the given spectral decomposition of A to compute a spectral decomposition of A^s , where s is any positive integer greater than 1.

54. Use the given spectral decomposition of A to find a spectral decomposition of a matrix C such that $C^3 = A$.

For an $n \times n$ matrix B and a polynomial $g(t) = a_n t^n + a_{n-1} t^{n-1} + \dots + a_1 t + a_0$, define the matrix $g(B)$ by

$$g(B) = a_n B^n + a_{n-1} B^{n-1} + \dots + a_1 B + a_0 I_n$$

55. Prove a special case of the Cayley-Hamilton theorem. If f is the characteristic polynomial of a symmetric matrix A , then $f(A) = 0$.

A is dia. , f cha poly , $f(A) = \mathbf{0}$

$A = A^T$, $A = \underline{P \Lambda P^T}$

$f(A) = \mathbf{0}$

43. Let $\mathbf{u}$ be a unit vector in R^n , and let P be the matrix $\mathbf{u}\mathbf{u}^T$. Prove that P is the orthogonal projection matrix for the subspace $\text{Span}\{\mathbf{u}\}$.